

DreaMS-conditioned Hybrid MS2LDA: Classical Topic Discovery with Semi-Amortized Spectral Inference

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Abstract

MS2LDA discovers recurring fragment and neutral-loss patterns by treating tandem mass spectra as documents in latent Dirichlet allocation (LDA). This note specifies a two-stage hybrid. First, sparse classical variational LDA discovers free topics by full-corpus expected-count updates, assisted by a bounded DreaMS-conditioned topic-word prior. Second, topic discovery is frozen permanently and a DreaMS document encoder is trained with a semi-amortized local LDA objective: the exact local evidence lower bound (ELBO) is differentiated through two unrolled coordinate updates. The encoder is never trained or used inside topic discovery. The implementation exposes a small Tomotopy-shaped interface, a one-way finalization lifecycle, and weights-only inference checkpoints. Focused tests and fresh-encoder frozen-topic experiments support a narrow limited-budget inference claim. Comparative motif discovery and chemical motif quality remain untested.

1 Introduction

Latent Dirichlet allocation represents a document as a mixture of latent topics and each topic as a probability distribution over words [1]. MS2LDA transfers this construction to tandem mass spectrometry: spectra are documents, discretized fragments and neutral losses are words, and topics are Mass2Motifs that may represent recurring molecular substructures [3]. One spectrum can contain several motifs, and one motif can recur across otherwise different molecules.

There are two distinct tasks. In *discovery*, the topic-word distributions are learned from a corpus. In *decomposition*, a new spectrum is assigned mixture weights over already learned topics. A network that predicts membership in fixed motifs can accelerate decomposition, but it cannot discover new motifs. A fully neural topic model can learn both parts, but changes the optimizer and can suffer component collapse or unstable topic identification [4].

This reference takes a middle route. Free topics are updated from variational LDA expected counts, while a frozen public spectral encoder supplies continuous features to an empirical-Bayes word prior and, after discovery, a semi-amortized new-document initializer. The two neural uses are deliberately separated: pooled peak features can bias the bounded word prior during discovery, whereas global spectrum embeddings train only the final local inference network. The scientific question is whether these features add useful chemical structure without replacing LDA’s auditable topic update. That question remains to be answered by the planned benchmark; this document first defines exactly what will be tested.

The contributions are:

1. a frozen DreaMS feature extractor that returns aligned spectrum-level and contextual peak representations;
2. one hybrid LDA model with classical topic discovery followed by a required semi-amortized inference-finalization stage;

3. a deliberately small adapter surface for fitting, inference, topic access, and safe inference-only checkpoints; and
4. a reproducible protocol for later comparison with Tomotopy and the frozen DreaMS representation itself.

2 Background

2.1 Classical and neural topic models

Mean-field variational inference for LDA alternates local document updates with global expected-count updates [1, 2]. Tomotopy uses a collapsed Gibbs sampler rather than mean-field variational inference, but fits the same broad LDA factorization. Both methods are stochastic and label-invariant: different seeds can find different local solutions even when all topics are used.

Neural topic models such as ProdLDA replace repeated local optimization with an amortized encoder [4]. This makes inference fast and allows continuous covariates, but creates an amortization gap: one shared mapping need not return the best variational posterior for every document [5]. Semi-amortized inference reduces this gap by using the encoder output as the start of local optimization [6]. Our model uses this idea while retaining a conjugate LDA topic posterior.

2.2 Pretrained DreaMS representations

DreaMS is a transformer pretrained by self-supervision on millions of unannotated tandem mass spectra [7]. It emits 1,024-dimensional contextual representations for spectral peaks and a spectrum-level representation at its precursor token. Its training objective and attention architecture encode peak context and pairwise mass differences, information absent from an MS2LDA count vector. We use the public `DreaMS_embedding` checkpoint without fine-tuning the transformer.

3 Method

3.1 LDA core

Let x_{dv} be the count of feature v in spectrum d , with K motifs and V retained features. Conditional on the empirically estimated $\boldsymbol{\eta}$, the model is classical LDA:

$$\boldsymbol{\beta}_k \sim \text{Dirichlet}(\boldsymbol{\eta}_k), \quad (1)$$

$$\boldsymbol{\theta}_d \sim \text{Dirichlet}(\boldsymbol{\alpha}), \quad (2)$$

$$z_{dn} \sim \text{Categorical}(\boldsymbol{\theta}_d), \quad (3)$$

$$w_{dn} \sim \text{Categorical}(\boldsymbol{\beta}_{z_{dn}}). \quad (4)$$

The topic-word variational posterior is $q(\boldsymbol{\beta}_k) = \text{Dirichlet}(\boldsymbol{\lambda}_k)$, and the document posterior is $q(\boldsymbol{\theta}_d) = \text{Dirichlet}(\boldsymbol{\gamma}_d)$. For each nonzero feature count,

$$\phi_{dvk} \propto \exp \left[\psi(\gamma_{dk}) - \psi \left(\sum_j \gamma_{dj} \right) + \psi(\lambda_{kv}) - \psi \left(\sum_u \lambda_{ku} \right) \right], \quad (5)$$

$$\gamma_{dk} = \alpha_k + \sum_v x_{dv} \phi_{dvk}, \quad (6)$$

$$\lambda_{kv} = \eta_{kv} + \sum_d x_{dv} \phi_{dvk}. \quad (7)$$

The last equation is a full-corpus expected-count update. The neural components do not replace it.

The implementation names the sparse count matrix `x`, the local parameters `gamma`, the responsibilities `phi`, and the global parameters `lambda_posterior`. The method `beta_mean()` returns the posterior mean of β . These names intentionally follow the equations.

3.2 Frozen spectrum and peak features

For spectrum d , the frozen DreaMS checkpoint produces a global representation $\mathbf{s}_d \in \mathbb{R}^{1024}$ and a contextual representation $\mathbf{h}_{dp} \in \mathbb{R}^{1024}$ for each retained peak p . The extractor uses the official DreaMS preprocessor and its top 100 peaks.

Contextual word features are constructed from training spectra only. For a fragment word at mass m_v , we pool the state of the nearest peak within 0.02 Da. For a neutral-loss word at loss ℓ_v , we pool the nearest peak to precursor mass minus ℓ_v . Repeated occurrences contribute according to their counts. This gives a mean contextual vector $\bar{\mathbf{h}}_v$. Unmatched words receive a zero contextual vector but retain their mass and word-type features.

3.3 Chemically structured topic–word prior

The contextual vector, numeric mass, and feature type are combined into a trainable normalized word representation:

$$\mathbf{u}_v = \text{normalize} \left[I_v W_h \text{LN}(\bar{\mathbf{h}}_v) + g_m(\tilde{m}_v) + \mathbf{e}_{\text{type}(v)} \right]. \quad (8)$$

Here $I_v \in \{0, 1\}$ records whether contextual peak evidence was observed, so an unmatched word contributes no contextual projection. The scalar input is $\tilde{m}_v = \log(1 + \max(m_v, 0)) / \log(2001)$. Here g_m is a small multilayer perceptron, and $\mathbf{e}_{\text{type}(v)}$ distinguishes fragments, losses, and other tokens. Each topic has a trainable normalized vector $\mathbf{t}_k$. The prior for topic k is

$$p_{kv} = \text{softmax}_v \left(\frac{\mathbf{t}_k^\top \mathbf{u}_v}{T} \right), \quad (9)$$

$$\eta_{kv}^{(e)} = \eta + r_e \rho \frac{N}{K} p_{kv}, \quad (10)$$

where N is the number of training tokens. The configuration defaults are $T = 0.5$, $\rho = 0.05$, and $r_e = \min(e/15, 1)$, giving a 15-epoch warm-up. Every topic therefore receives the same structured prior mass.

With row-normalized topic vectors collected in $\tilde{T}$, the prior parameters minimize

$$-\frac{1}{KV} \sum_k \mathbb{E}_q[\log \text{Dir}(\beta_k; \boldsymbol{\eta}_k^{(e)})] + 10^{-3} \frac{\|\tilde{T} \tilde{T}^\top - I\|_F^2}{K^2}. \quad (11)$$

With the default configuration, one Adam update is taken per global epoch through epoch 20 and the prior is then fixed. Automatic discovery convergence is declared only when the relative L_1 change in $\boldsymbol{\lambda}$ is below 10^{-3} for three consecutive fixed-prior epochs; 100 epochs is the default maximum. These are configurable operational rules, not a proof of ELBO convergence. A caller may deliberately stop discovery and finalize after any completed epoch, accepting the risk of locking a premature topic state.

3.4 DreaMS-conditioned document encoder

The encoder is anchored to the final free topics. Define

$$\ell_{kv} = \psi(\lambda_{kv}) - \psi\left(\sum_u \lambda_{ku}\right), \quad (12)$$

$$r_{vk} = \text{softmax}_k(\ell_{kv}), \quad (13)$$

$$e_{dk} = \sum_v \frac{x_{dv}}{N_d} r_{vk}, \quad N_d = \sum_v x_{dv}. \quad (14)$$

For an empty in-vocabulary document, $\mathbf{e}_d$ is defined as the uniform topic vector. The spectrum representation follows the exact projection

$$\mathbf{p}_d = \text{LN}_2(\text{GELU}[W_s \text{LN}_1(\mathbf{s}_d) + \mathbf{b}_s]), \quad (15)$$

which has 128 output dimensions by default. It is concatenated with $\mathbf{e}_d$ and passed through two 256-unit ReLU hidden layers. The output is a topic-logit residual:

$$\mathbf{a}_d = g_\omega[\mathbf{e}_d; \mathbf{p}_d], \quad (16)$$

$$\hat{\boldsymbol{\pi}}_d = \text{softmax}(\log[\max(\mathbf{e}_d, \epsilon)] + \mathbf{a}_d), \quad (17)$$

$$\boldsymbol{\gamma}_d^{(0)} = \boldsymbol{\alpha} + N_d \hat{\boldsymbol{\pi}}_d. \quad (18)$$

The final residual layer starts at zero, so the untrained network begins from topic evidence supplied by LDA rather than an unrelated coordinate system. The construction also preserves $\sum_k \gamma_{dk}^{(0)} = \sum_k \alpha_k + N_d$ exactly.

Training VB starts from the retained $\boldsymbol{\gamma}_d$ from the preceding epoch, not from the encoder. Consequently, global spectrum embeddings affect new-document inference but not discovered topics. Only pooled contextual peak features affect discovery, through the structured word prior. This is an explicit separation of classical discovery and neural local inference, not a ProgLDA-style end-to-end topic model. The encoder and spectrum projector are neither evaluated nor optimized during discovery.

During discovery, document posteriors are refined with up to 50 sparse local coordinate updates and tolerance 10^{-4} before the global expected-count update. For a new spectrum, the encoder returns an immediate estimate, which may be followed by standard local LDA updates. Refinement changes only that document’s topic mixture; it does not change learned motifs.

3.5 Required semi-amortized inference finalization

After the final topic update, `finalize_inference()` performs the required second stage exactly once. The complete topic posterior $\boldsymbol{\lambda}$ and all structured-prior parameters are treated as fixed, and a fresh Adam optimizer updates only the document encoder and spectrum projector for 12 epochs by default. The implementation snapshots and checks the topic posterior and every trainable structured-prior parameter bit-for-bit before declaring finalization successful. If optimization or a progress callback fails, it restores the encoder, random-generator state, history, and module mode. Topic discovery cannot resume after successful finalization, and new-document inference and checkpoint saving are unavailable beforehand. If a discovery progress callback finalizes the just-completed epoch, the outer discovery loop detects the one-way state transition and stops before another topic update.

For a candidate document posterior $\boldsymbol{\gamma}_d$, write

$$\bar{\theta}_{dk} = \psi(\gamma_{dk}) - \psi\left(\sum_j \gamma_{dj}\right). \quad (19)$$

With the categorical responsibilities set to their coordinate optimum, the encoder-dependent local sparse LDA ELBO is compactly

$$\mathcal{L}_d(\gamma_d; \lambda) = -\text{KL}[\text{Dir}(\gamma_d) \parallel \text{Dir}(\alpha)] + \sum_v x_{dv} \text{LSE}_k(\bar{\theta}_{dk}(\gamma_d) + \ell_{kv}). \quad (20)$$

Written out, up to constants that do not depend on document inference, this is

$$\begin{aligned} \mathcal{L}_d(\gamma_d; \lambda) = & \log \Gamma(\alpha_0) - \sum_k \log \Gamma(\alpha_k) - \log \Gamma(\gamma_{d0}) + \sum_k \log \Gamma(\gamma_{dk}) \\ & + \sum_k (\alpha_k - \gamma_{dk}) \bar{\theta}_{dk} + \sum_v x_{dv} \log \sum_k \exp(\bar{\theta}_{dk} + \ell_{kv}), \end{aligned} \quad (21)$$

where $\alpha_0 = \sum_k \alpha_k$ and $\gamma_{d0} = \sum_k \gamma_{dk}$. The final term is exactly the usual expected assignment term plus categorical entropy, evaluated with a stable log-sum-exp.

Starting from the encoder output $\gamma_d^{(0)}$, two fixed coordinate steps are unrolled:

$$\phi_{dvk}^{(t)} = \text{softmax}_k(\bar{\theta}_{dk}^{(t)} + \ell_{kv}), \quad (22)$$

$$\gamma_{dk}^{(t+1)} = \alpha_k + \sum_v x_{dv} \phi_{dvk}^{(t)}. \quad (23)$$

Equivalently, these equations define a differentiable map $F_d(\gamma)$ with $\gamma_d^{(1)} = F_d(\gamma_d^{(0)})$ and $\gamma_d^{(2)} = F_d(\gamma_d^{(1)})$. Back-propagation passes through both maps and into $\gamma_d^{(0)}$; it does not pass into λ or the structured prior. The single supported encoder objective is

$$\mathcal{J}_{\text{SA}} = -\frac{\sum_d [\mathcal{L}_d(\gamma_d^{(2)}; \lambda) + 0.1 \mathcal{L}_d(\gamma_d^{(0)}; \lambda)]}{\sum_d N_d}. \quad (24)$$

For a uniformly shuffled minibatch of B documents from a corpus of D documents, its ELBO sum is multiplied by $D/(B \sum_d N_d)$. This is the correctly scaled uniform-document minibatch estimator of Equation 24. The implementation uses random reshuffling, Adam, and gradient-norm clipping; later minibatches in an epoch therefore see updated parameters and should not be described as independent draws. Normalizing separately by each random minibatch’s token total would instead reweight short and long batches. Epoch progress values are online minibatch diagnostics because the encoder changes between updates, not a fresh evaluation of the final epoch parameters. The small zero-step term retains useful encoder-only predictions. Training uses exactly two updates with no adaptive stopping inside the differentiable graph, making the objective and its back-propagation path fixed for every document.

At inference time, the default is the same two-update budget, although a caller may request another exact number of local updates. Alternatively, a positive tolerance makes that number a maximum batch-level budget and stops when $\max_{d,k} |\gamma_{dk}^{(t+1)} - \gamma_{dk}^{(t)}| / \max(|\gamma_{dk}^{(t)}|, \epsilon)$ falls below the threshold. Adaptive stopping is never used in the differentiable training graph. Reported training-document topic mixtures are also recomputed from retained classical-VB state under the final topics, rather than from the neural initializer.

3.6 Training and inference sequence

The implementation follows one explicit sequence:

1. build a training-only vocabulary and sparse document–word matrix, then initialize the topic posterior from a seeded random simplex;

2. refine each training document’s γ_d and ϕ_{dv} using the current topics;
3. accumulate full-corpus expected topic–word counts, update the bounded structured prior during its declared window, and set λ to prior plus expected counts;
4. continue until the caller stops or stationary-prior convergence is declared;
5. permanently freeze the final topics and train the encoder through two differentiable local updates using Equation 24; and
6. for an unseen spectrum, compute the encoder initialization and optionally apply local VB without changing λ .

No dense document-by-vocabulary tensor is constructed. Each batch contains only padded nonzero word identifiers and counts.

3.7 Implementation and compatibility boundary

Conceptually the system is one fitted `HybridLDAModel` plus one frozen external feature extractor. `MS2LDA/hybrid_lda.py` contains the sparse LDA equations, neural modules, global update, checkpoint, and adapter. `MS2LDA/dreams_features.py` contains the lazy DreaMS loader, aligned HDF5 cache, and train-only peak-to-word pooling. There is no second model implementation.

`HybridLDAModel` implements only the small subset of `tomotopy.LDAModel` needed by model-facing experiments: adding documents, classical discovery, inference finalization, topic access, new-document inference, document topic distributions, likelihood summaries, and save/load. It is not wired into the existing CLI, dashboard, or visualizer. Tomotopy remains the production backend.

The version-3 reference checkpoint can be written only after successful finalization. It contains the topic posterior, neural parameters, configuration, vocabulary, and explicit finalized state. It deliberately excludes training documents, local variational state, optimizer state, and random-number-generator state and is therefore inference-only. It is loaded with PyTorch’s weights-only mode. Feature provenance is stored in the DreaMS cache rather than the model checkpoint, so callers must keep compatible artifacts together.

4 Validation status

The focused tests verify configuration invariants, sparse count handling, the local LDA updates and ELBO, monotonic coordinate refinement, differentiable two-step refinement, conservation of token and Dirichlet mass, fixed prior mass, stationary-prior convergence, deterministic same-seed discovery and finalization, absence of encoder updates during discovery, bitwise frozen-topic and prior invariance during finalization, the one-way lifecycle, adaptive inference, strict weights-only checkpoint round trips, feature caching, peak-to-word pooling, and the shared spectral-word parser.

4.1 Frozen-topic inference experiments

The fixed objective was first explored in a deterministic oracle-topic simulator and then rerun on seeds 23, 29, and 31 from fresh seeded encoders. A benchmark-only posterior-mean regression comparator received the same topics, initial encoder, documents, minibatch order, and 12 training epochs. The common per-document reference was the maximum ELBO among 100-step solves from the symmetric initializer and both trained encoder initializers, followed by one diagnostic continuation step. The semi-amortized objective reduced the two-step ELBO gap by 13.37%, 20.04%, and 14.15%, respectively. Their arithmetic mean was 15.86%; the relative reduction between the across-seed mean gaps was

15.84%. Across-seed mean encoder-only observed-token NLL rose by 0.086%, while two-step NLL was slightly lower. This simulator isolates inference behavior; its oracle topics and latent-derived embeddings are not chemical evidence.

A second experiment used the archived random-split mushroom artifacts only to freeze the same discovered topics and reconstruct the exact split: 2,171 training spectra, 539 eligible test spectra, 10,029 training-vocabulary words, 209,497 observed test tokens, and 214,271 held-out test tokens. For each topic seed 42–46, both comparison encoders were reset to the same fresh seeded initialization before any final-stage training. Each received the same frozen topics, document order, minibatch random-number-generator state, and 12 epochs. The benchmark comparator regressed to normalized local-VB posterior means; the current method optimized Equation 24. Test embeddings came from partial spectra containing observed physical peaks only.

The semi-amortized arm reduced the two-step signed ELBO gap in all five seeds: 7.39–10.52%, with mean 9.53% and sample standard deviation 1.38 percentage points. Three of five seeds exceeded a prespecified 10% diagnostic threshold; the threshold was retained rather than revised after observing the rerun. Held-out NLL was lower in all five seeds at both zero and two refinement steps. The mean improvements were 0.0481 nats/token at zero steps and 0.0146 at two steps; mean two-step NLL was 7.0048 rather than 7.0194. The current method at two steps was also 0.0499 nats/token better than the comparator after five steps. Every topic and structured-prior tensor remained bitwise unchanged.

Long local solves were diagnostics, not claims of uniform numerical convergence. Across seeds, one additional coordinate update was positive for as many as 107 documents and had a maximum individual gain of 0.9063 nats. The largest corpus-average continuation gain was only 1.02×10^{-5} per observed token, too small to explain the finite-budget differences. As an artifact-integrity check, migrated predictions reproduced stored NLL values with mean absolute error 1.28×10^{-8} and maximum absolute error 3.71×10^{-8} nats/token.

The synthetic driver is `scripts/benchmark_semi_amortized_inference.py`; the trusted-artifact mushroom driver is `scripts/benchmark_mushroom_inference_phase.py`. The regression comparator lives only in `scripts/inference_baselines.py`; it is not exposed by the production model or serializable as a finalized checkpoint. Both drivers retain machine-readable raw results. These experiments support a narrow claim: from a fresh seeded encoder and fixed topics, the final semi-amortized encoder is a better limited-budget document-inference initializer than this comparator. They do not show that discovered motifs are more chemically meaningful, more stable, or superior to another topic-discovery method.

5 Planned topic-discovery and chemical benchmark

5.1 Artifacts and split discipline

The benchmark will live outside the reference implementation on a validation branch. It must contain a single documented driver, a redistributable input or an immutable input manifest, exact train/test assignments, configuration and seed records, and machine-readable outputs. Generated feature caches and model checkpoints may remain external when large, but their hashes and construction commands must be recorded.

Vocabulary construction and pooled word features must use training spectra only. Two 80/20 evaluations will be frozen before results are inspected:

1. a seeded random spectrum split; and
2. a grouped split that keeps DreaMS-similar spectra on the same side.

For document completion, fragment and neutral-loss words originating from the same physical peak must be kept together. A test-spectrum DreaMS embedding must be recomputed from its observed peak groups rather than from the complete spectrum.

5.2 Compared representations

The primary topic-model comparison is the current hybrid against the existing Tomotopy LDA backend, trained from scratch with the same topic count and input counts over seeds 42–46. A raw global-DreaMS similarity baseline is also required. It is not a competing topic model; it tests whether mapping the pretrained representation through topics adds information beyond using that representation directly.

Training budgets and inference settings must be stated explicitly. Report both model-side latency with cached DreaMS features and end-to-end latency that includes feature extraction. A latency-matched Tomotopy setting should accompany its standard inference setting.

5.3 Metrics and claims

Predictive quality will use held-out negative log likelihood per token:

$$\text{NLL} = -\frac{1}{N_{\text{held}}} \sum_{d,v} x_{dv}^{\text{held}} \log \left(\sum_k \hat{\theta}_{dk} \hat{\beta}_{kv} \right). \quad (25)$$

Lower NLL is better. Topic diagnostics will include active topics, top-word diversity, word-co-occurrence NPML, and cross-seed stability after optimal topic matching. Same-seed matched cosine and top-word Jaccard similarity will describe overlap between methods, not establish correctness.

Chemical evaluation must use labels withheld from training and model selection. The preferred primary endpoint is recovery of existing manual motif–spectrum annotations on an independent, redistributable corpus. Spectral-family retrieval may be reported as supporting evidence, with family-macro average precision and paired uncertainty intervals, but spectral network components are not direct substructure annotations.

Metrics, exclusions, seed count, and the primary endpoint must be fixed before test results are inspected. All seeds must be reported. Results from this protocol should replace the validation-status section rather than be combined with the inference-only exploratory results above.

6 Limitations

The hybrid retains LDA’s non-convex, label-invariant topic discovery, so different seeds can still reach different solutions. DreaMS is trained to represent whole-spectrum molecular context, whereas a Mass2Motif is intended to capture a smaller recurring pattern. A useful global representation does not therefore guarantee a useful motif prior.

The structured-prior mass, warm-up, and temperature are fixed design choices that have not yet been validated. The two-step objective and 0.1 zero-step weight were selected in an internal simulator; the fresh-encoder mushroom rerun is a useful robustness check but not an independent model-selection study. Finalization intentionally locks topic discovery, so fitting on a premature topic state cannot be repaired without training a new model. The model is not an end-to-end neural topic model, and the production MS2LDA workflow does not yet select it as a backend. Finally, a passing software test suite and frozen-topic inference improvement cannot replace external chemical validation.

7 Reproducibility

The self-contained reference surface consists of `MS2LDA/hybrid_lda.py`, `MS2LDA/dreams_features.py`, `tests/test_hybrid_lda.py`, `scripts/inference_baselines.py`, the two benchmark drivers named above, `environment-hybrid.yml`, and this document. A small lazy import in `MS2LDA/__init__.py` prevents the full application stack from loading when the reference modules are imported.

The environment uses Python 3.11 and pins NumPy 1.25.0, PyTorch 2.2.1, and DreaMS at Git commit `dbec3a0b514a99e5056cfccde4559fda8cfe8129`. Its required architecture package is pinned at `d35b34b0caee38348098cda852bff30d8b25cd25`. Installing DreaMS with `--no-deps` is deliberate: the YAML supplies the packages imported by the embedding path and prevents DreaMS’s moving `maml@main` requirement from replacing the exact architecture commit.

The reference can be installed and tested with:

```
conda env create -f environment-hybrid.yml
conda activate ms2lda-hybrid
python -m pip install --no-deps \
    "git+https://github.com/pluskal-lab/DreaMS.git@dbec3a0b514a99e5056cfccde4559fda8cfe8129"
python -m pip install --no-deps -e .
python -m pytest -q tests/test_hybrid_lda.py
```

DreaMS caches record the expected package commit, installed package version, both public checkpoint hashes, embedding dimension, and peak limit. Model loading does not enforce a cache match. Reference checkpoints contain no spectra and cannot resume training.

8 Conclusion

DreaMS-conditioned Hybrid MS2LDA is a focused reference implementation of conjugate free-topic discovery followed by semi-amortized local inference. The mathematical ownership is explicit: expected counts discover topics, the bounded structured prior biases their word distributions, and only after those quantities are frozen does the document encoder learn through the local LDA ELBO. Frozen-topic experiments from fresh encoders support the limited-budget inference mechanism, but they do not justify replacing Tomotopy for motif discovery or claiming chemically superior Mass2Motifs.

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